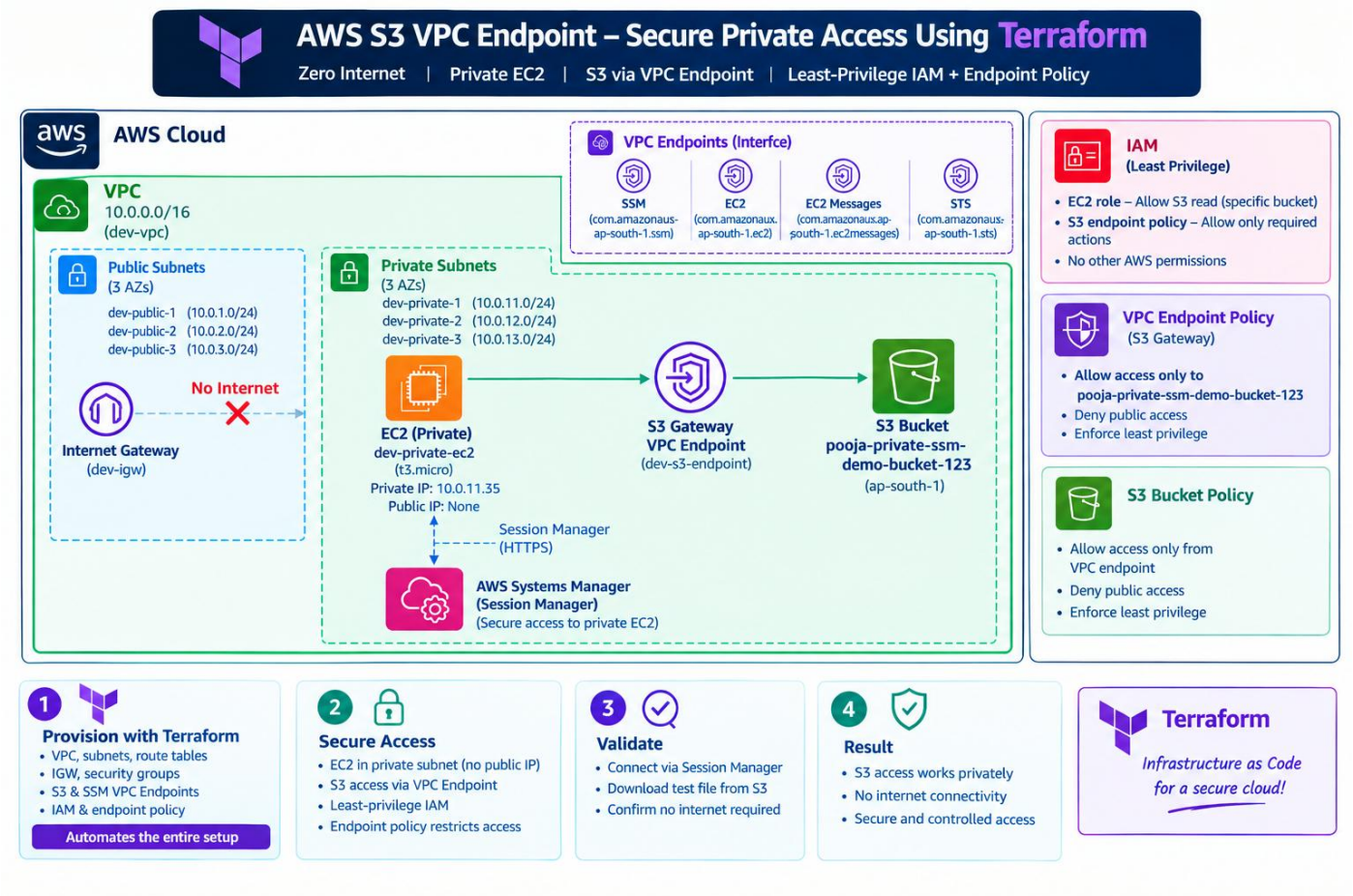
AWS S3 VPC Endpoint – Secure Private Access Using Terraform

Prepared by: Pooja Daingade

Project Overview

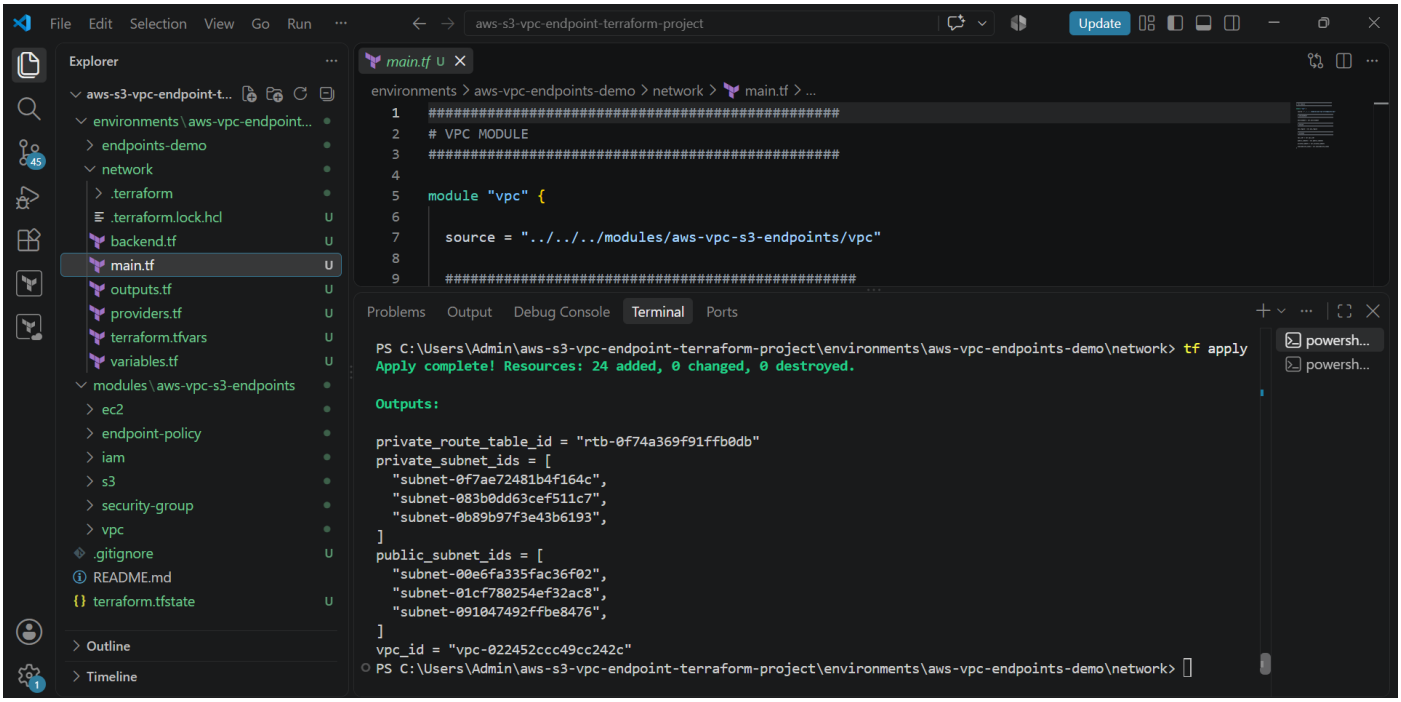
This project focuses on building a secure, zero-internet AWS architecture using Terraform. An EC2 instance runs in a private subnet without a public IP address and accesses Amazon S3 through an S3 Gateway VPC Endpoint. Least-privilege IAM permissions and an endpoint policy restrict access to authorized resources. AWS Systems Manager Session Manager enables secure instance access, while downloading a test file from S3 verifies that private connectivity works without internet access.

Architecture Diagram



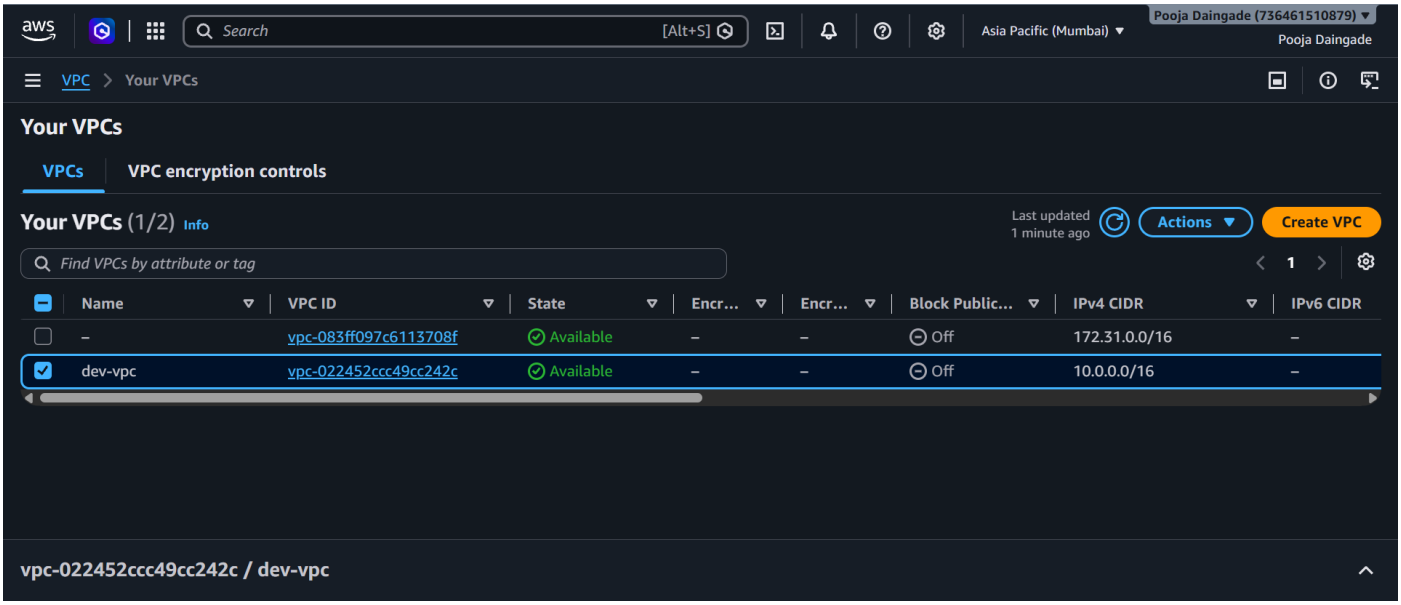
1. VPC Module Deployment Using Terraform.

The screenshot shows the Terraform project in VS Code and the successful deployment of the VPC infrastructure using the tf apply command.

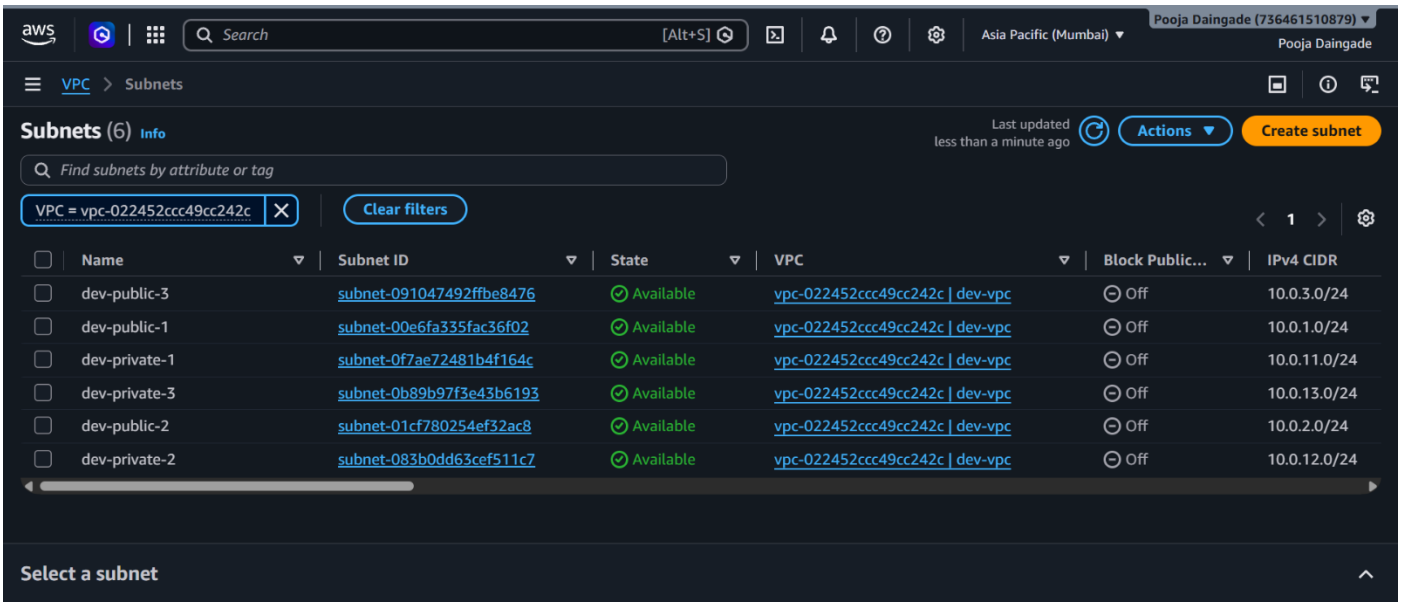


- **VPC Infrastructure Deployment**

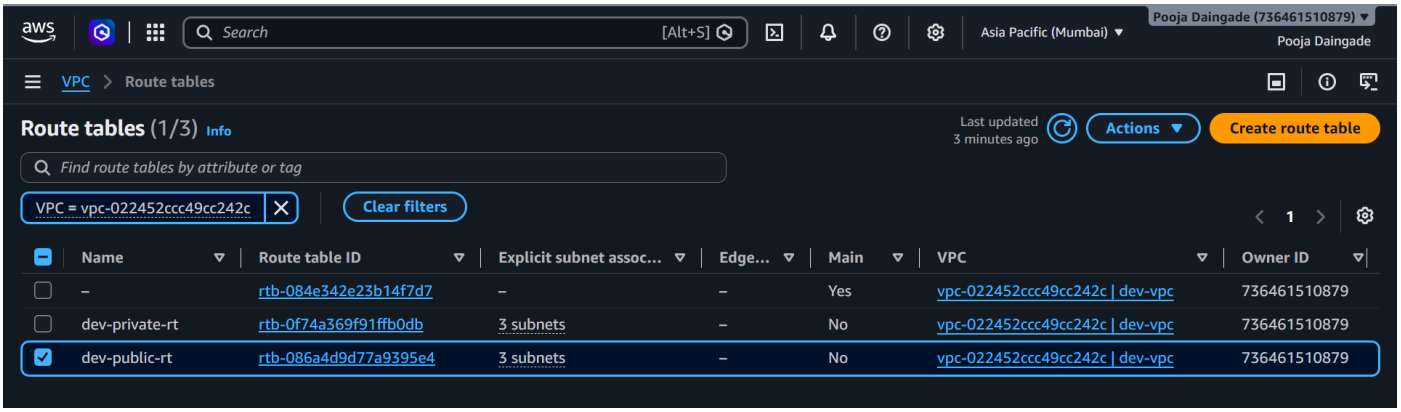
  The AWS VPC Console shows the successfully created **dev-vpc** with CIDR block **10.0.0.0/16** and status **Available**.

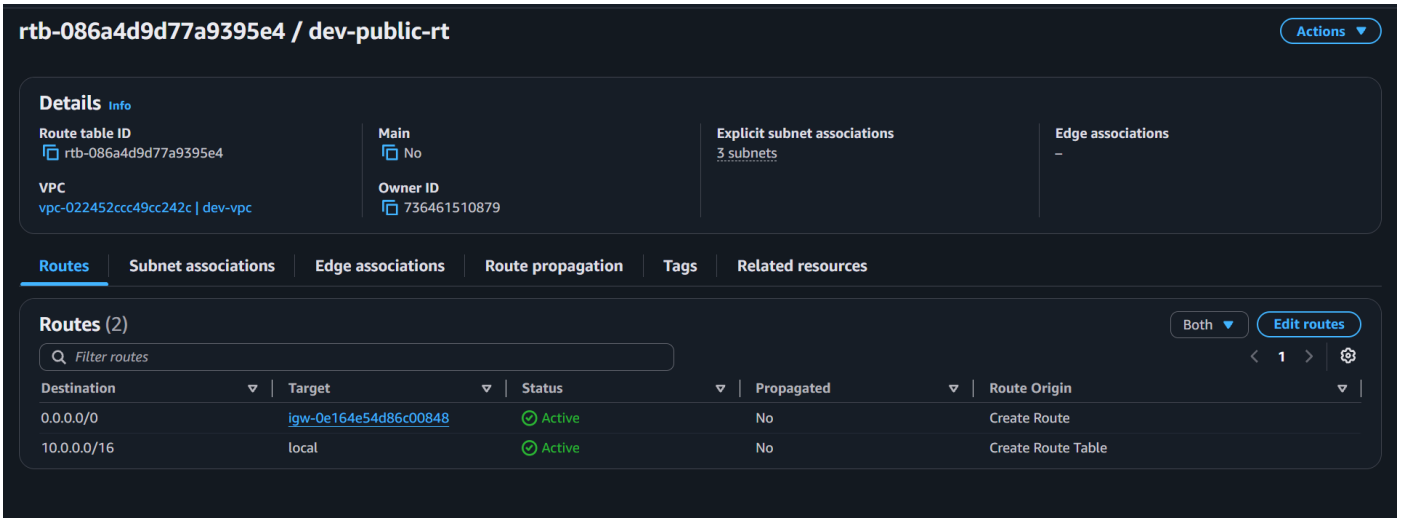


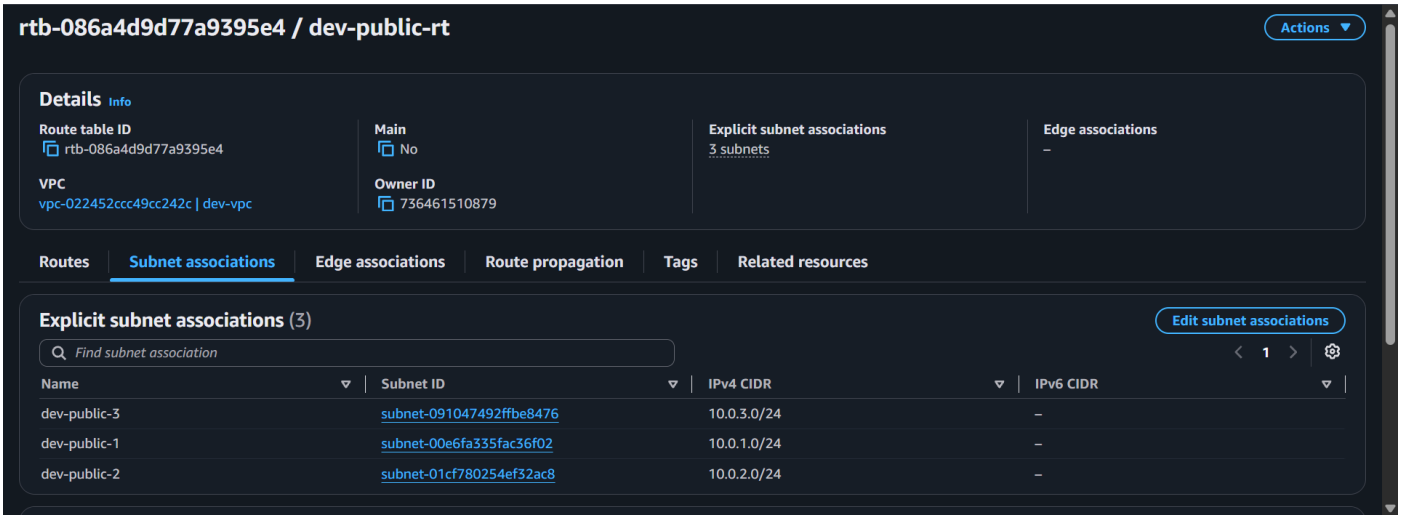
- **Public and Private Subnets**
  The AWS VPC Console shows **six subnets** successfully created inside the dev-vpc VPC.
  Public Subnets: dev-public-1, dev-public-2, and dev-public-3
  Private Subnets: dev-private-1, dev-private-2, and dev-private-3



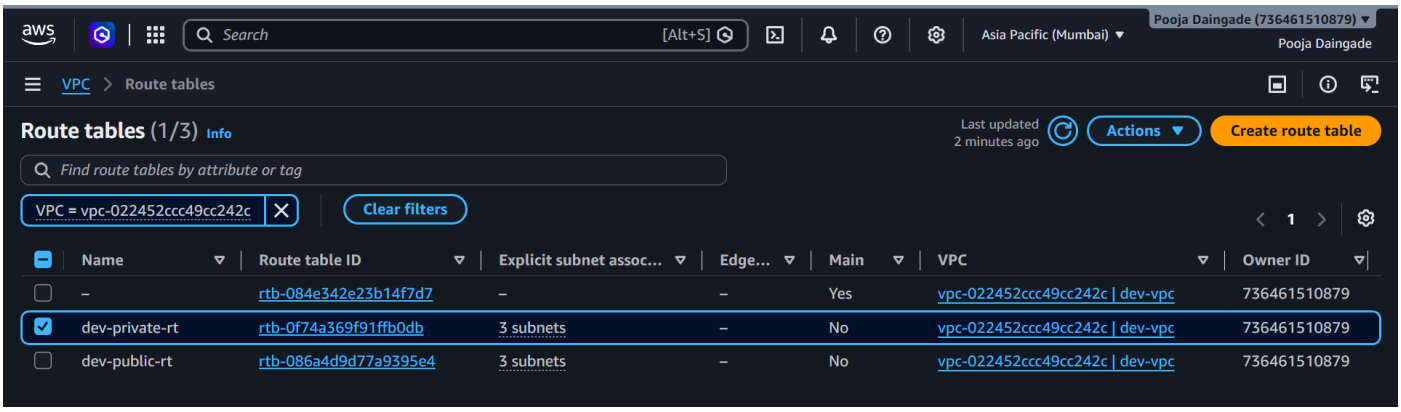
- **Public and Private Route Tables**
  The AWS Console shows two custom route tables configured for dev-vpc:
  dev-public-rt — Associated with 3 public subnets.

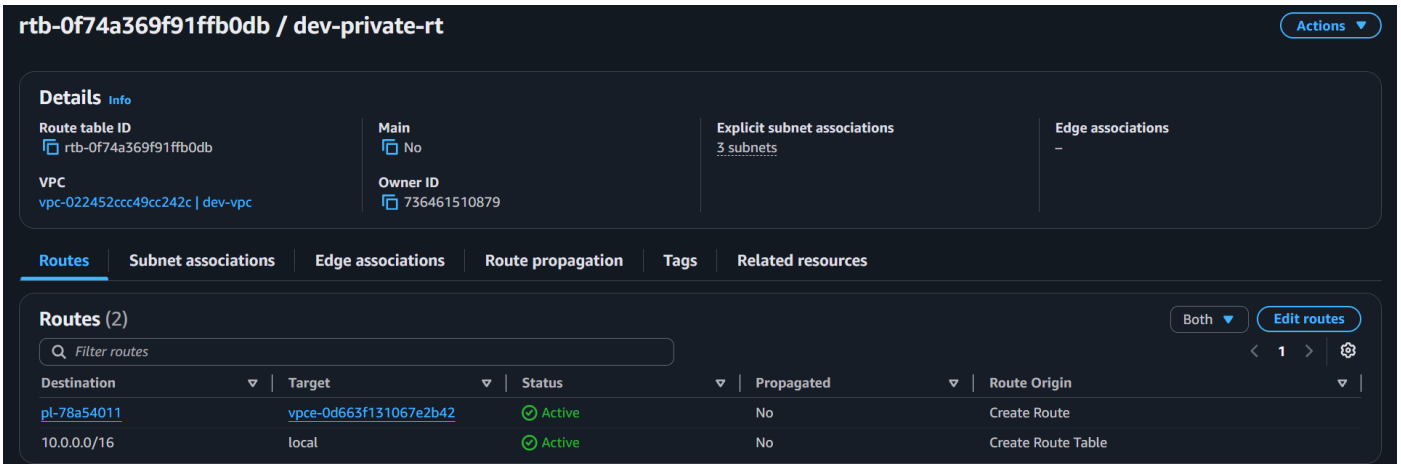


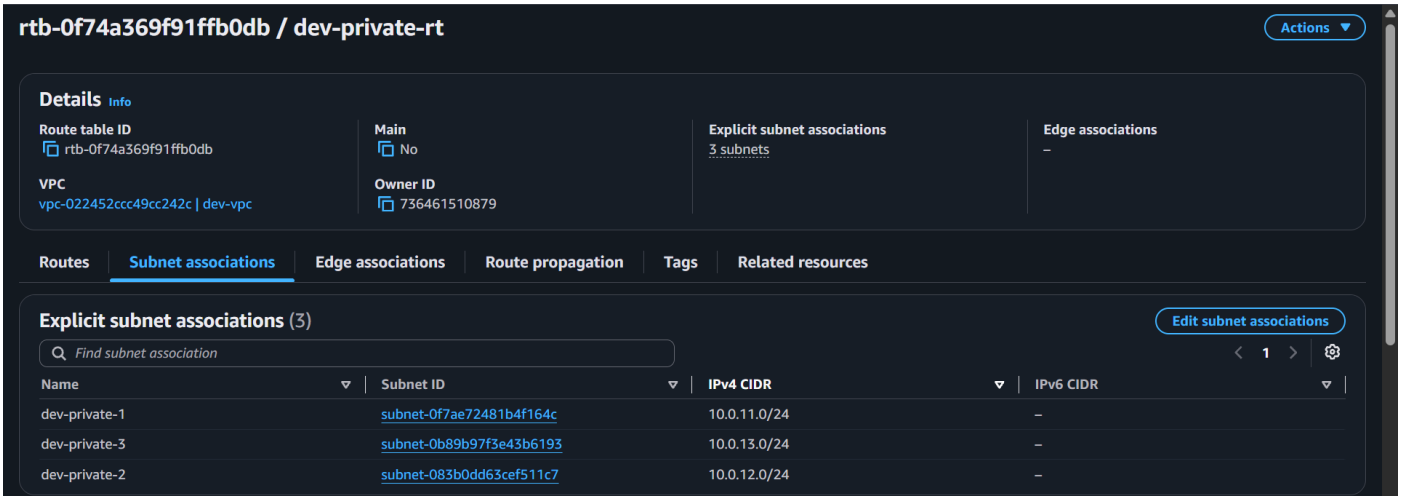




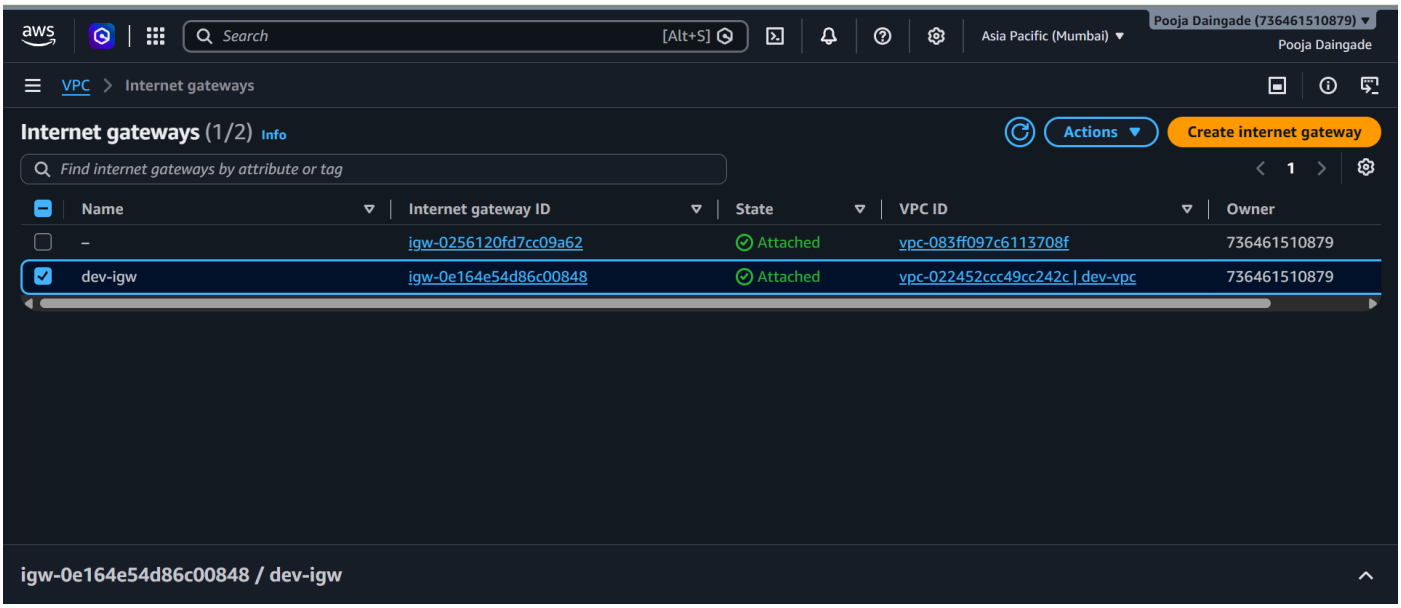
dev-private-rt — Associated with 3 private subnets.



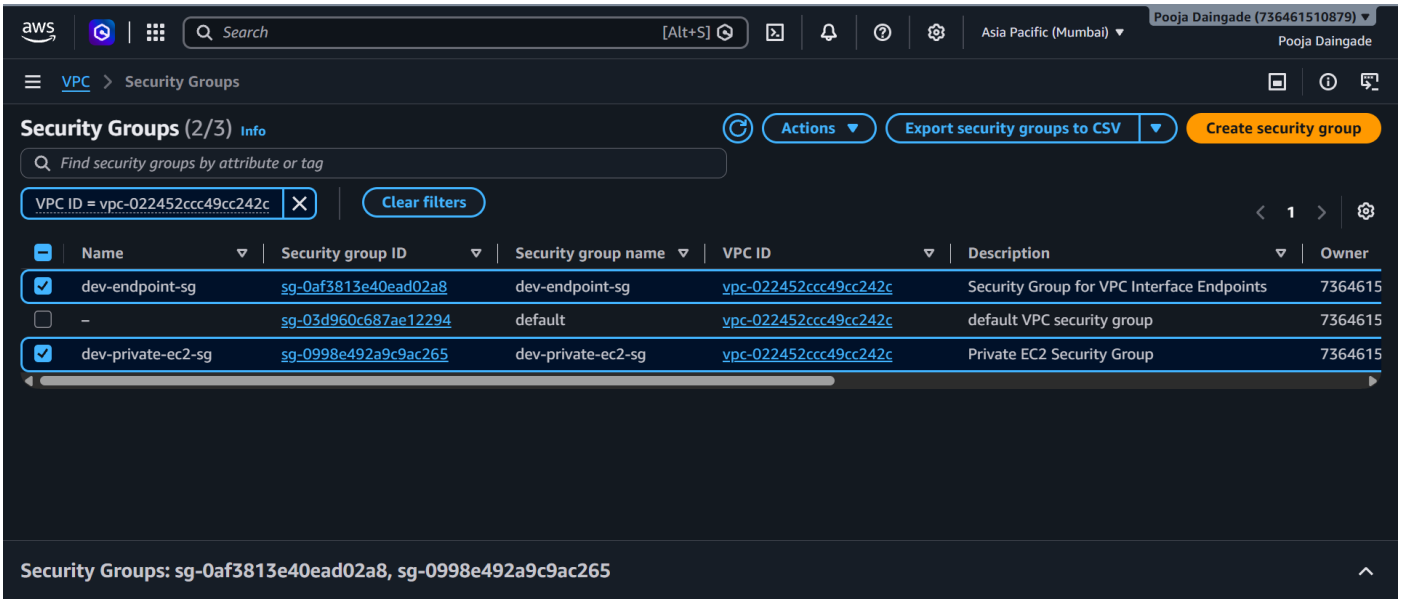




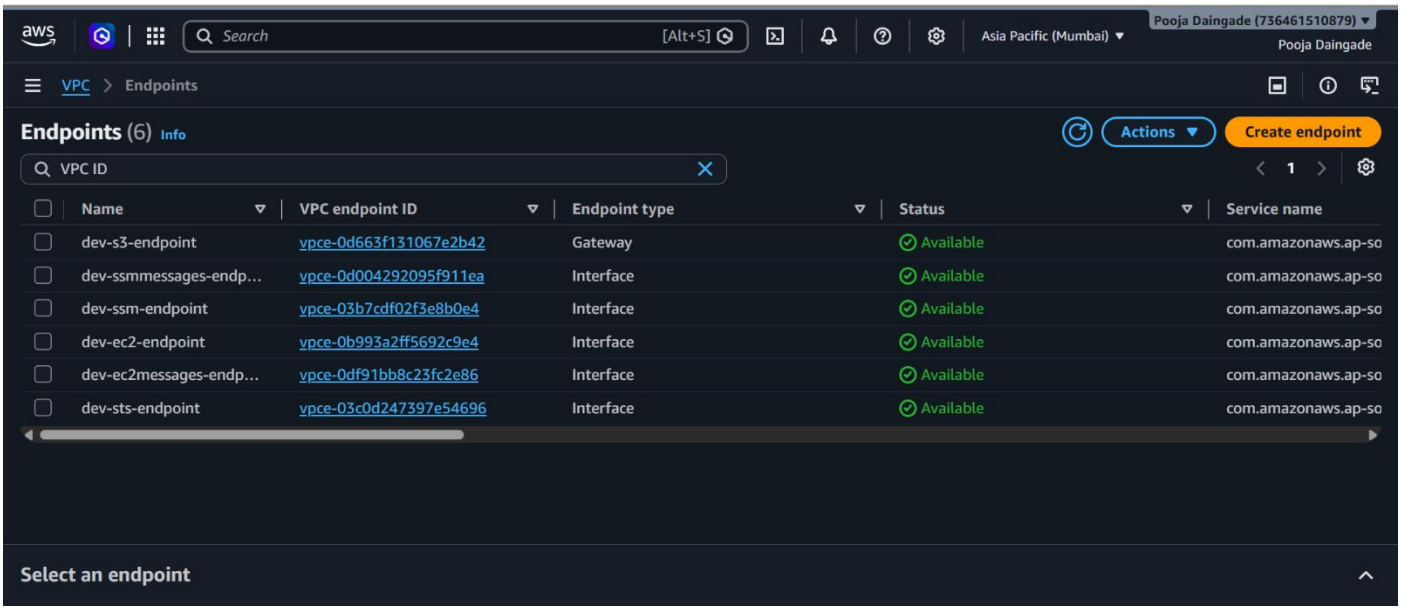
- **Internet Gateway Creation and Attachment**
  The Internet Gateway **dev-igw** was created and attached to the dev-vpc VPC.



- **Security Group Configuration**
  The AWS Console displays two custom security groups created in the dev-vpc VPC:
  dev-private-ec2-sg – Controls network traffic for private EC2 instances.
  dev-endpoint-sg – Controls traffic for VPC interface endpoints.

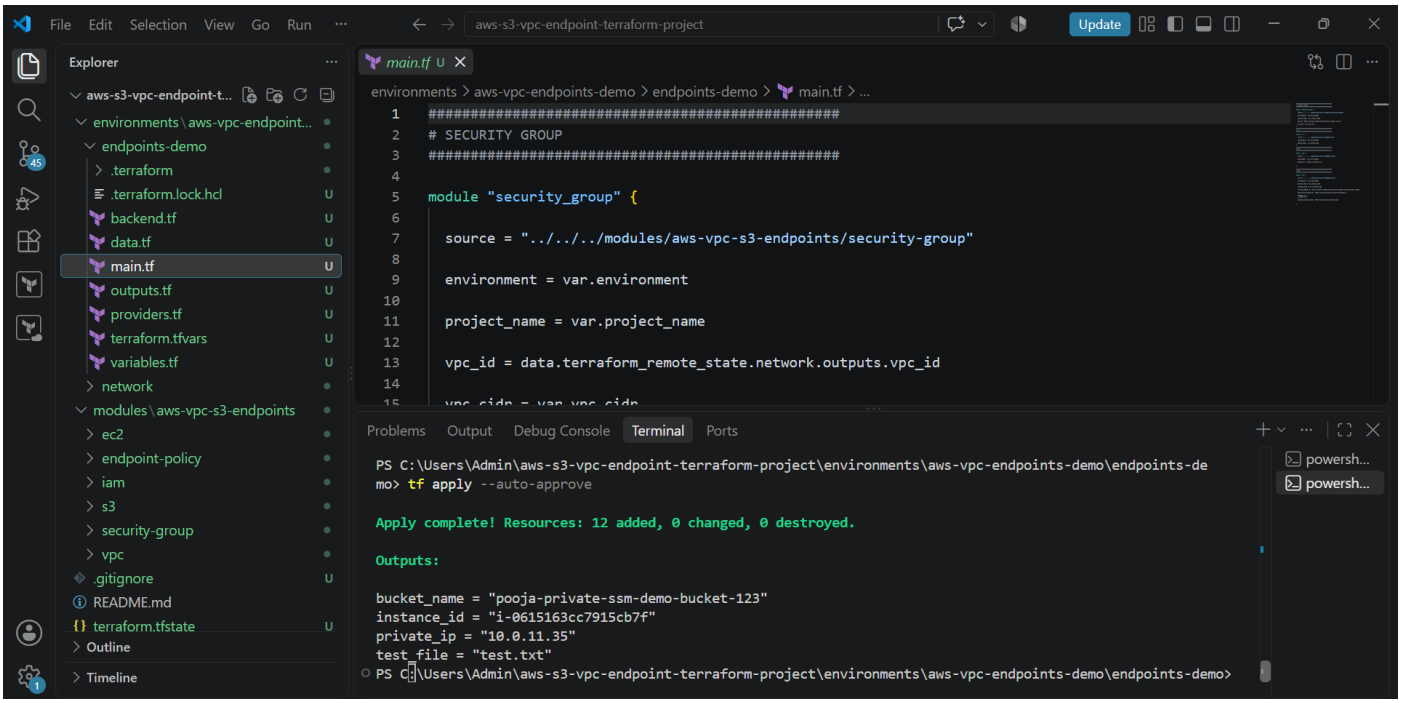


- **VPC Endpoint Configuration**
  The AWS Console shows six VPC endpoints successfully created, all with **Available** status.
  S3 Gateway Endpoint: dev-s3-endpoint provides private access to Amazon S3.
  SSM Interface Endpoints: Support secure EC2 management through AWS Systems Manager Session Manager.
  Additional Endpoints: EC2, EC2 Messages, and STS endpoints support required AWS service communication.

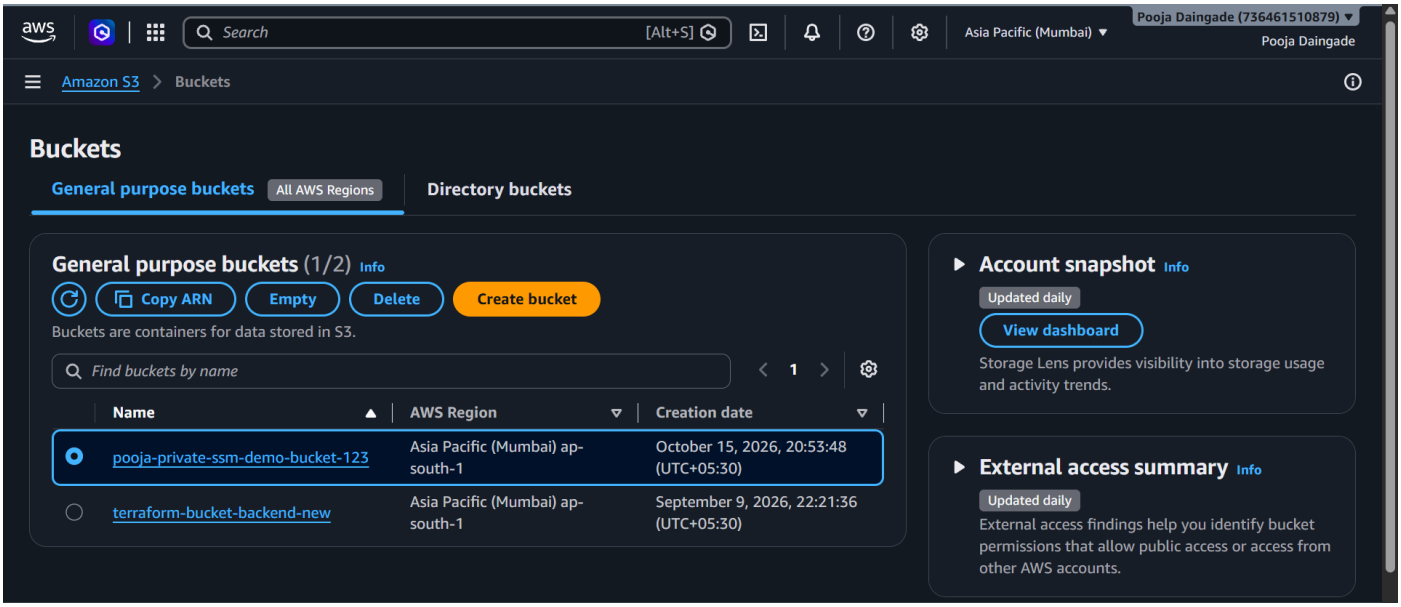


2. Private EC2 and S3 Infrastructure Deployment

The infrastructure was successfully provisioned using Terraform for testing private EC2 access to Amazon S3.

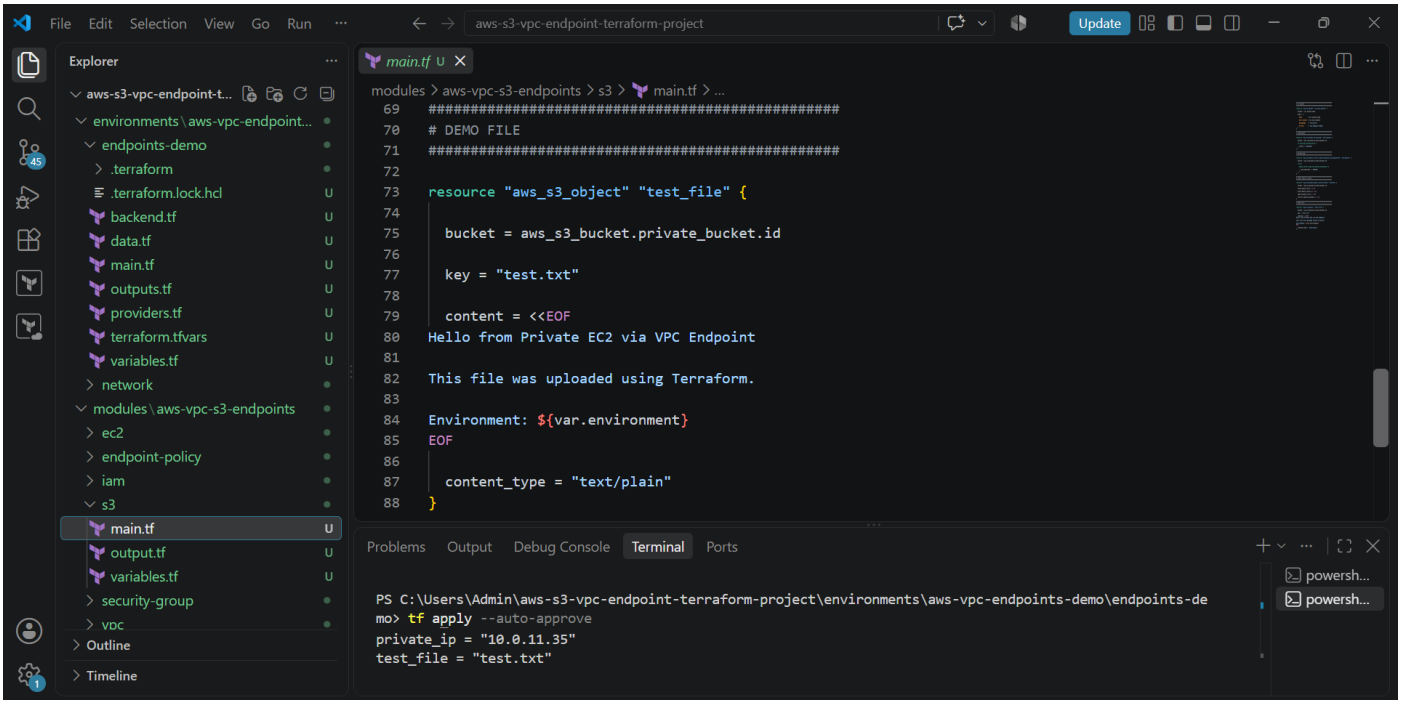


- **Amazon S3 Bucket Verification**
  The AWS S3 Console confirms the successful creation of the bucket **pooja-private-ssm-demo-bucket-123** in the Mumbai region (ap-south-1).

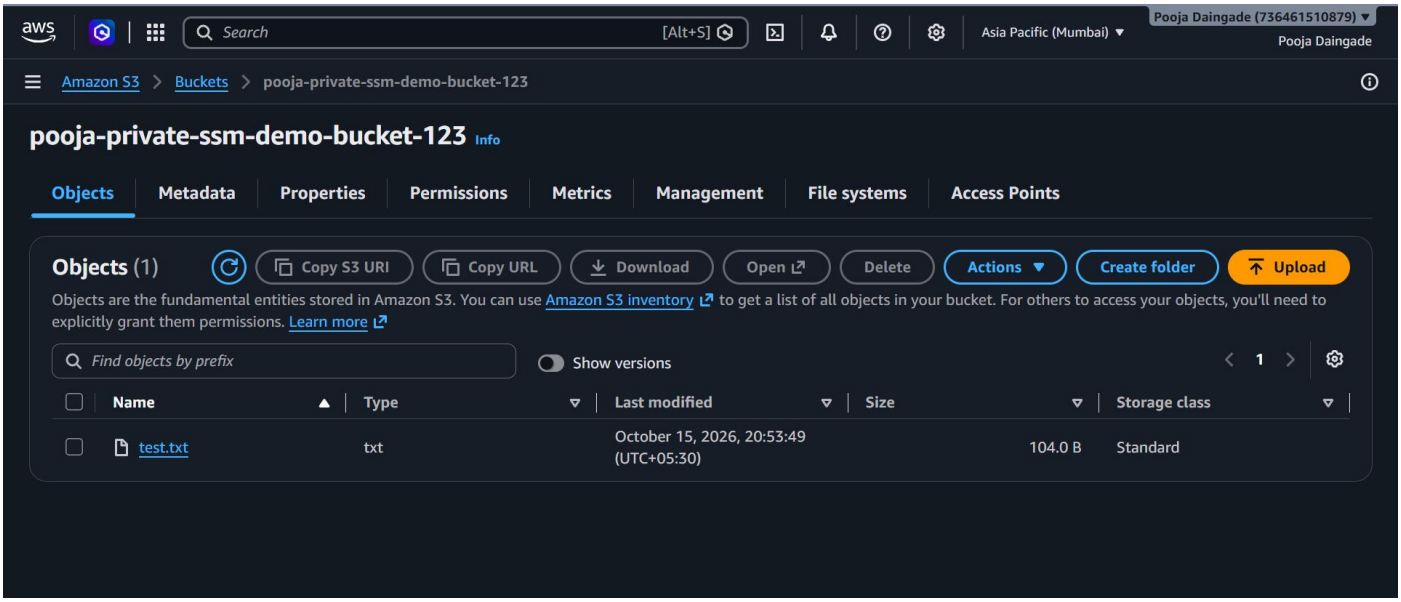


3. Upload Test File to Amazon S3 Using Terraform

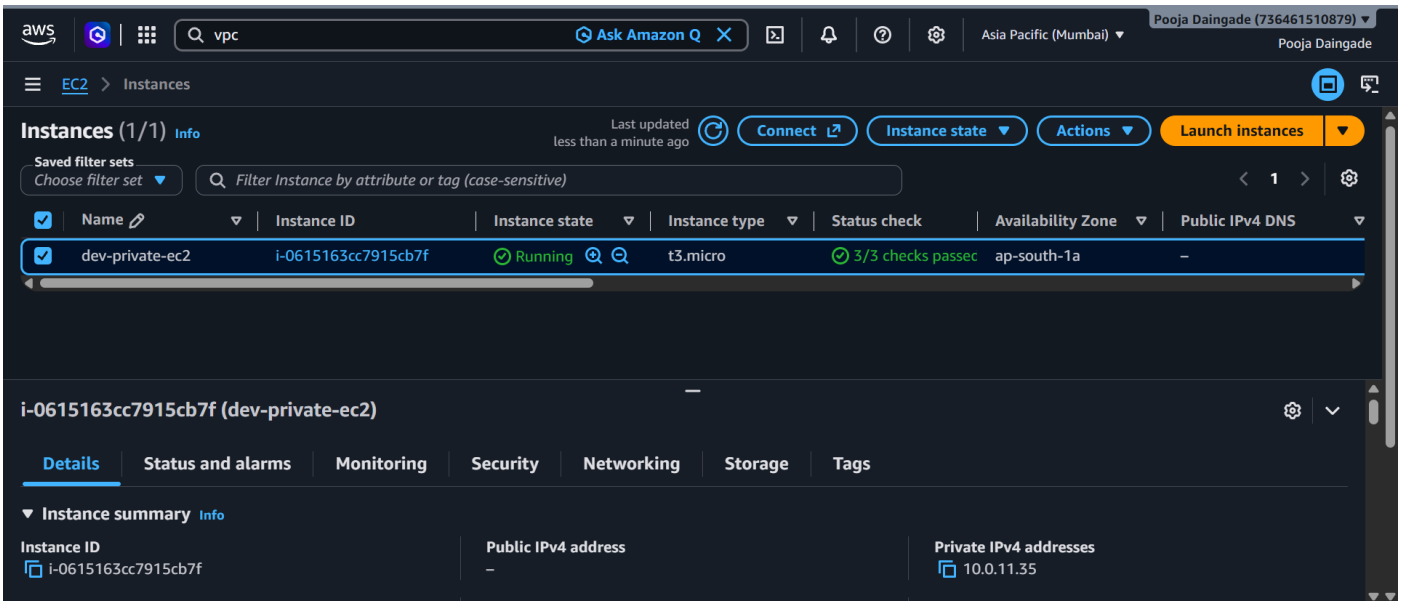
The Terraform S3 module defines the test.txt object with sample content and uploads it to the private S3 bucket.



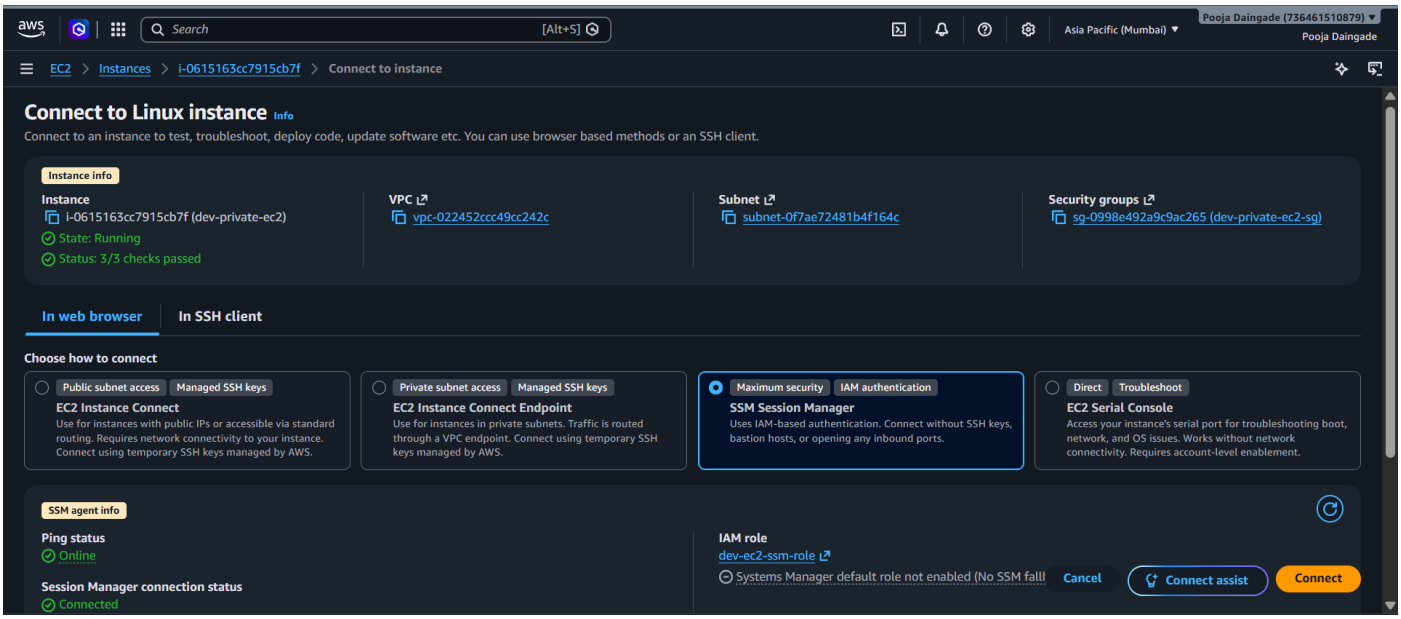
- **Verify test.txt in the S3 Bucket**
  The AWS S3 Console displays the test.txt object inside the private S3 bucket, confirming that the file was uploaded successfully using Terraform.



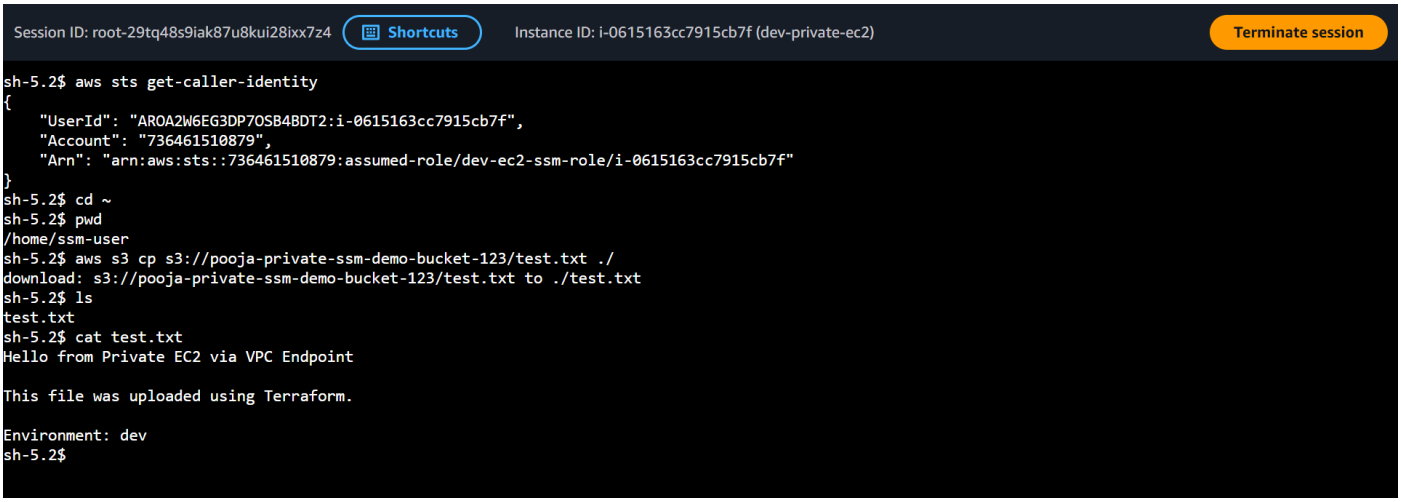
- **Private EC2 Instance Verification**
  The EC2 **dev-private-ec2** instance is running in a private network without a public IP address, supporting secure access through AWS Systems Manager Session Manager.
  Instance Type: t3.micro
  Private IP: 10.0.11.35
  Public IP: None
  Availability Zone: ap-south-1a



- **Connecting to Private EC2 Using AWS SSM**
  The private EC2 instance (dev-private-ec2) is running with private IP 10.0.11.35 and has passed all status checks. AWS Systems Manager Session Manager shows **Online** and **Connected**, enabling secure browser-based access without a public IP address or SSH keys.



- **Verify S3 Access from Private EC2**
  Using AWS Systems Manager Session Manager, the private EC2 instance downloads **test.txt** from the S3 bucket through the VPC Endpoint. The **cat test.txt** command confirms the file content was retrieved successfully without requiring a public IP address.



Conclusion

The project successfully demonstrates how to securely access Amazon S3 from a private EC2 instance without requiring internet connectivity. Terraform automates infrastructure deployment, while IAM permissions and the VPC Endpoint policy strengthen access control. The successful retrieval of a test file from S3 validates private connectivity and demonstrates a secure approach to AWS resource access.